

The Editors
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Dear Editors,

We submit for your consideration our manuscript, “A versioned, leakage-audited USGS *Did You Feel It?* benchmark for machine-learning evaluation of severe felt intensity.” We request handling as a Research Article documenting a public seismological data/software benchmark resource.

Machine-learning studies of macroseismic felt intensity commonly use bespoke datasets and random record-level splits. Their results can be difficult to reproduce and vulnerable to leakage between related earthquake events. Our manuscript addresses this problem by releasing a frozen, single-source USGS DYFI benchmark, a six-category leakage audit, outcome-independent sequence-grouped roles, fixed baselines, calibration-aware scoring, and a protected temporal evaluation.

The frozen $M \geq 5$ snapshot contains 7,520 source records, 2,362 eligible events, and a 430-event temporal holdout. The observed random-versus-sequence split differences are small and reported descriptively; no winning model, operational forecasting benefit, causal effect, or cross-system generalization is claimed. All deviations and unexecuted branches are disclosed.

The work fits the *Journal of Seismology* as an observational macroseismology and citizen-science data/software resource. It directly aligns with the journal’s stated interest in machine learning, seismological datasets, software tools, and publicly available resources supporting transparent, reproducible earthquake science.

Data, code, frozen results, and reproduction instructions are available at <https://github.com/aksha74/dyfi-benchmark-reliability>. A clean clone reproduces the 16-page manuscript, passes the released manuscript validator and benchmark test suite, and reconstructs the recorded results and holdout predictions. The same tagged release contains a non-peer-reviewed manuscript PDF and source; we disclose this as a preprint under Springer Nature’s preprint policy. The preprint has no DOI. No separate licence is granted for the manuscript text; copyright remains with the authors pending publication (the released software and data retain their stated MIT and CC0 terms). We will update the record with the article DOI and journal URL if published.

The manuscript is original, is not under consideration elsewhere, and has not been published previously. The authors declare no competing interests. The study uses aggregate public-domain event metadata, involves no human participants, and redistributes no respondent-level identifying information. Generative-AI assistance is disclosed in Methods and all assisted work was verified by the authors.

Sincerely,

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